

AFTERZERO: THE LAST VERSION OF EARTH

COMPLETE 25–30 HOUR GAMEPLAY SCRIPT

A cinematic temporal-reality thriller combining third-person action, stealth, open-region exploration, causal puzzles, alternate histories, and emotionally consequential choices.

FORMAT	CAMPAIGN TARGET	STRUCTURE
Single-player narrative action-adventure	25–30 hours	23 missions + optional cases

Design note: durations are targets, not measured playtest results. This document is a narrative/gameplay blueprint for production planning; it does not represent a completed game.

1. THE GAME AT A GLANCE



Logline

In 2049, temporal systems engineer Kai Varen activates a machine that lets him interact with different states of the same physical world. Searching for his sister, Mira, he discovers that every attempt to repair history has helped create a planetary catastrophe—and that the person guiding him may be the future version of himself.

Creative pillars

• Temporal cause and effect: actions in one era alter routes, evidence, hazards, factions, and character outcomes in another.

• Mystery with fair clues: twists should feel surprising on first viewing and logical in hindsight.

• Cinematic action with quieter investigation: combat, traversal, environmental puzzles, and character scenes alternate.

• Consequences without punishment: major choices change missions and relationships but do not casually block the main campaign.

• A personal ending: the final decision is about grief, identity, and the limits of control—not simply defeating a villain.

Campaign timing

Section	Target	Core experience
Prologue — The Zero Second	2 h	First anomaly; tutorial; Chronolith
Act I — The World That Remembers	5 h	Discovery; ancient mystery; causal puzzles
Act II — The Nine-Minute War	6 h	Combat escalation; looping city; factions
Act III — The Last Version of Earth	6 h	Future exploration; consequences; Witness
Act IV — The Unwritten Second	5 h	Unreliable memory; reality-breaking puzzles
Finale and epilogue	1.5 h	Final choice; ending; sequel hook

2. WORLD, CHARACTERS AND CORE SYSTEMS



Main characters

Kai Varen — protagonist. A temporal systems engineer who believes every event has a discoverable cause. His flaw is believing every loss can be reversed.

Mira Varen — the impossible sister. Kai's sister, officially dead for 23 years. She is a reconstructed consciousness carrying memories from more than one possible life.

The Witness — older Kai. A future Kai who has repeatedly tried to save Mira, creating new timelines and guiding his younger self toward the same experiment.

Sera Voss — resistance leader. A survivor from 2196 who distrusts Kai because he is remembered as the architect of the Fracture.

Dr. Elias Rook. Director of the Helix project; hides the early history of the Chronolith.

Niko. A young technician whose clues appear across multiple temporal states.

Dr. Anja Vale. A future scientist who studies temporal memory and the human cost of erased timelines.

The four temporal states

• Past: reveals original structures, causes, and objects that no longer exist.

• Present: supports investigation, modern machinery, grounded stealth, and human conflict.

• Future: exposes consequences, enables traversal through ruins, and introduces unstable temporal zones.

• Inverted state: temporarily reverses selected processes, subject to energy and stability limits.

Signature systems

• Temporal State Switching — move between authored states of one location; not unrestricted date selection.

• Temporal Residue — inspect objects for visual/audio traces of previous events.

• Causal Anchors — selected objects persist between states after a deliberate action.

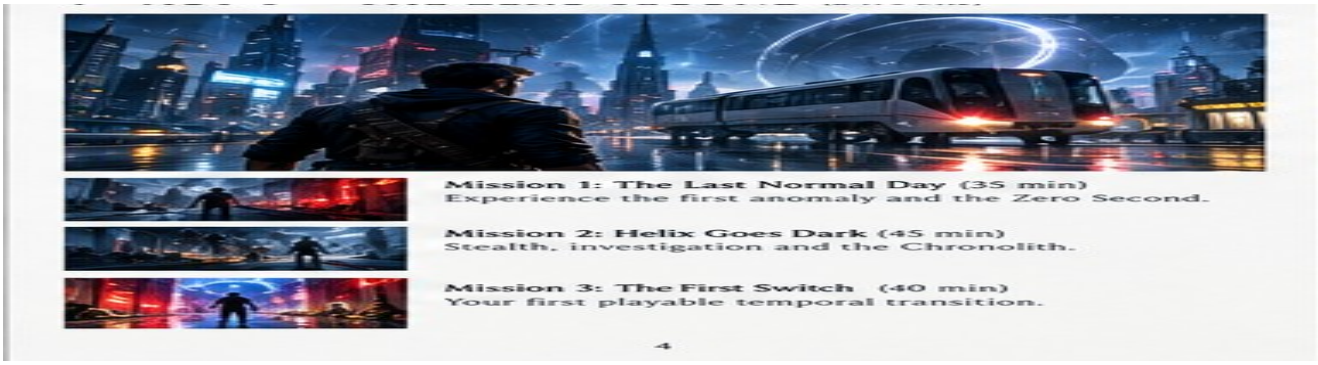
• Echo Step — brief afterimage that distracts enemies or crosses a dangerous gap; consumes stability.

• Local Loops — observe repeated sequences and use knowledge to solve timing-based puzzles.

• Persistent Consequences — large actions modify the condition of locations in later missions.

> *Design rule: every major temporal puzzle must be solvable through observable evidence. The player may be surprised, but solutions must not feel arbitrary.*

3. PROLOGUE — THE ZERO SECOND (2 HOURS)



MISSION 1: THE LAST NORMAL DAY

Target duration: 35 minutes **Purpose:** Introduce Kai and establish the first anomaly.

Story setup — Kai commutes through a rain-soaked city to Helix Temporal Research Facility. Small impossible events occur around him: a toy jumps back into a child's hand, a clock runs backward, and a camera displays tomorrow's footage.

Playable sequence

• Explore the city and inspect optional anomalies.

• Receive an emergency message from Dr. Rook.

• Reach the facility through a crowded security checkpoint.

• When all clocks stop, rescue trapped civilians as a train crashes into the platform.

> *""The event lasted one second." / "What event?" / "That's the problem. Every recording says it hasn't happened yet.""*

Mission ending: A voice speaks through every speaker: "Kai Varen. Do not build the machine." Kai sees a woman resembling Mira vanish into the crowd.

MISSION 2: HELIX GOES DARK

Target duration: 45 minutes **Purpose:** Introduce stealth, residue investigation and the Chronolith.

Story setup — The facility locks down. Rook calls the Zero Second a system failure but blocks access to the central chamber.

Playable sequence

• Navigate dark corridors using emergency lighting.

• Avoid malfunctioning security drones.

• Find three power cells and restore a lift.

• Use recorded conversations to identify a trustworthy officer.

• Fight a security unit that rewinds damage; lure it into a damaged power conduit.

> *""This drawing is twenty-three years old." / "Then someone has been using your credentials." / "I was twelve.""*

Mission ending: Kai finds a child's drawing of himself and Mira beside a black circular machine, then enters the Chronolith chamber.

MISSION 3: THE FIRST SWITCH

Target duration: 40 minutes **Purpose:** Teach the temporal state-switching loop.

Story setup — The facility cycles between its operating state in 2049, unfinished state in 2026, and ruins in 2196.

Playable sequence

• Switch to 2026 and connect a power cable.

• Return to 2049 and activate the power.

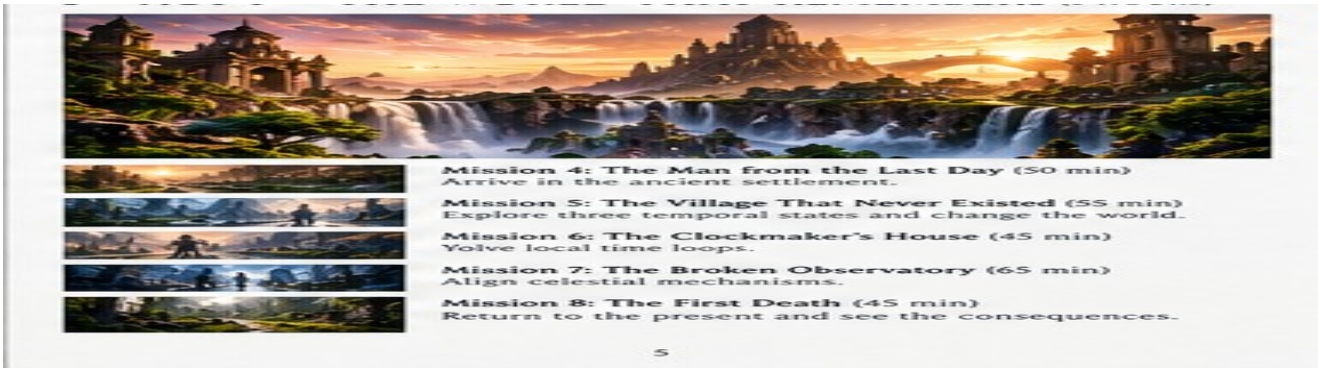
• Switch to 2196 to use a lift over collapsed machinery.

• Return to 2049 and reach the control panel.

> *“Which one is real?” / “All of them. That’s why this is dangerous.”*

Mission ending: Kai is pulled into the past as the facility fractures.

4. ACT I — THE WORLD THAT REMEMBERS (5 HOURS)



MISSION 4: THE MAN FROM THE LAST DAY

Target duration: 50 minutes **Purpose:** Establish the ancient mystery.

Story setup — Kai wakes beside a settlement built around a stone structure resembling the Chronolith. Its leader recognizes a symbol on his wrist.

Playable sequence

• Explore the settlement and learn its visual language.

• Find three inscription fragments.

• Follow clues to an underground chamber.

• Reconstruct the sequence of construction, prosperity, destruction and abandonment.

> *“‘You came before the first fire.” / “I’ve never been here.” / “That is what the others said.”’*

Mission ending: Inside the chamber is a Helix access card bearing Kai’s name, dated 3,000 years earlier.

MISSION 5: THE VILLAGE THAT NEVER EXISTED

Target duration: 55 minutes **Purpose:** Show persistent cross-era consequences.

Story setup — The village appears in three states: thriving, abandoned, and flooded. Niko follows Kai through transitions.

Playable sequence

• Explore all three versions of the village.

• Divert a river in the earliest state.

• Reinforce a retaining wall in the middle state.

• Release a pressure gate in the latest state.

• Discover a buried machine far more advanced than its era.

> *“‘You keep changing the world.” / “I’m trying to get home.” / “Then stop changing it.”’*

Mission ending: Niko reveals that Helix did not invent the machine; it found it.

MISSION 6: THE CLOCKMAKER’S HOUSE

Target duration: 45 minutes **Purpose:** Teach local temporal loops.

Story setup — A clockmaker claims Kai visited decades earlier. His workshop repeats short sequences of mechanical actions.

Playable sequence

• Observe a repeating mechanical arm and cross the workshop.

• Reveal a hidden staircase by comparing room states.

• Set clock mechanisms to different positions in past, present and future.

• Find a recording left by older Kai.

> *“‘You ask what time it is. Wrong question.” / “What’s the right one?” / “Which version of the question are you asking?”’*

Mission ending: A recording warns: “When the child returns, do not let him see the ninth minute.”

MISSION 7: THE BROKEN OBSERVATORY

Target duration: 65 minutes **Purpose:** Expand vertical exploration and celestial logic.

Story setup — An ancient observatory tracks celestial alignments and connects to the Chronolith.

Playable sequence

• Align three instruments using the sun and stars in different states.

• Traverse rooftops, rotating platforms and an exterior ring.

• Fight temporal scavengers whose attacks displace objects backward.

• Discover dates that span thousands of years, including Kai’s birth date.

> *“‘How could they know when I would be born?” / “They didn’t know when. They knew who.”’*

Mission ending: Mira’s hidden recording implies the Chronolith has been anticipating Kai, not merely recording him.

MISSION 8: THE FIRST DEATH

Target duration: 45 minutes **Purpose:** Reveal that the present has changed.

Story setup — Kai returns to a Helix facility whose layout and staff records have changed after his actions in the past.

Playable sequence

• Investigate an altered facility.

• Find a missing researcher.

• Recover a corrupted security log.

• Reconstruct the final ten minutes before the Zero Second.

• Escape a temporal containment chamber.

> *“‘Don’t save me.” / “Why?” / “Because I am not the one who died.”’*

Mission ending: The system identifies the current timeline as VERSION 17; no surviving record exists for the previous sixteen.

5. ACT II — THE NINE-MINUTE WAR (6 HOURS)



MISSION 9: NINE MINUTES TO ZERO

Target duration: 60 minutes **Purpose:** Introduce knowledge-based looping.

Story setup — A city repeats the same nine-minute period with changing enemy positions, civilian routes and damage.

Playable sequence

• Observe two loops and map enemy patrols.

• Find three synchronized clocks.

• Sabotage the central transmitter.

• Rescue a trapped engineer and reach the tower before reset.

> *“‘You have nine minutes.” / “To do what?” / “To become me.”’*

Mission ending: The city resets, but Kai retains his knowledge and no longer resets with it.

MISSION 10: ENEMIES WHO REMEMBER

Target duration: 65 minutes **Purpose:** Introduce faction conflict and Echo Step.

Story setup — The Continuum captures Kai, believing time must be controlled. Sera helps him escape.

Playable sequence

• Escape a detention block using stealth.

• Switch power systems across states to distract guards.

• Use a captured temporal weapon.

• Fight through a testing chamber.

• Choose whether to release prisoners trapped in unstable loops.

> *“‘In my time, your face is on every memorial.” / “For saving the world?” / “For ending it.”’*

Mission ending: Kai unlocks Echo Step, a limited-use temporal afterimage.

MISSION 11: THE INVERTED HOSPITAL

Target duration: 50 minutes **Purpose:** Deliver a tense non-combat puzzle mission.

Story setup — A hospital is caught in localized inversion: some rooms move forward while others reverse.

Playable sequence

• Follow sound that becomes quieter as Kai approaches.

• Avoid hazards returning to original positions.

• Compare medical records to identify a missing patient.

• Dismantle a reversing lock in the correct order.

> *“Temporal discontinuity.”*

Mission ending: Mira’s file lists an age of 32, though the record predates the present by 23 years; her death date is blank.

MISSION 12: THE MAN BEHIND THE GLASS

Target duration: 70 minutes **Purpose:** Reveal the Witness without resolving the mystery.

Story setup — Kai pursues a figure who changes states to escape.

Playable sequence

• Chase through Helix in 2049.

• Use a corridor that existed only in 2026.

• Traverse the ruined facility in 2196.

• Reconstruct the Witness’s route from repeated fragments.

• Enter a chamber containing a second Chronolith.

> *“The person you become when you refuse to let go.”*

Mission ending: The Witness bears the same wrist scar as Kai and escapes through a fracture.

MISSION 13: THE NINE-MINUTE WAR

Target duration: 75 minutes **Purpose:** Deliver the first large-scale battle and a consequential choice.

Story setup — Two factions fight over a stabilizer that can freeze a district or free a trapped population.

Playable sequence

• Cross an active battlefield.

• Disable three devices in different temporal states.

• Recognize that both factions are reacting to their own previous attacks.

• Choose to destroy the stabilizer or redirect its energy to free civilians.

> *“This is not the future.” / “What is it, then?” / “The future that survives long enough to remember us.”*

Mission ending: The choice changes the district and later support options, but does not block the campaign.

6. ACT III — THE LAST VERSION OF EARTH (6 HOURS)



MISSION 14: WELCOME TO 2196

Target duration: 50 minutes **Purpose:** Make the catastrophe tangible.

Story setup — Kai arrives in a beautiful, devastated world and must reach a settlement before the Chronolith loses stability.

Playable sequence

• Traverse overgrown skyscrapers and ruined transit lines.

• Avoid unstable temporal zones.

• Scavenge components to repair a return beacon.

• Track survivors and discover a memorial depicting Kai as the architect of the Fracture.

> *“We teach children your name before they learn the names of the dead.”*

Mission ending: Sera reveals that every version of Kai denies responsibility.

MISSION 15: THE CITY THAT FORGOT TOMORROW

Target duration: 65 minutes **Purpose:** Explore memory loss as a societal consequence.

Story setup — Residents repeat daily routines and cannot remember events beyond the current day.

Playable sequence

• Investigate a train station reliving an evacuation.

• Explore a school reliving its abandonment.

• Investigate a market reliving an undocumented disaster.

• Separate causes, consequences and fabricated evidence in a central archive.

> *“That depends on whether you count the worlds that never finished becoming worlds.”*

Mission ending: The player stabilizes the city or frees it from the loop; later allies and dialogue change.

MISSION 16: THE MUSEUM OF FAILED WORLDS

Target duration: 55 minutes **Purpose:** Humanize erased timelines.

Story setup — An underground museum stores objects from worlds that no longer exist.

Playable sequence

• Play three short memories from different versions of one event.

• Compare artifacts with the current world.

• Determine which version best explains the museum evidence.

• Find a photograph of older Kai and Mira.

> *“‘We had a thousand chances. We only needed one that was ours.’”*

Mission ending: The museum proves the Witness has repeated the experiment many times.

MISSION 17: THE WITNESS’S ARCHIVE

Target duration: 70 minutes **Purpose:** Explain the original tragedy and the Witness’s motive.

Story setup — The archive contains recordings from multiple versions of Kai’s life.

Playable sequence

• Navigate shifting corridors.

• Match recordings to physical evidence.

• Fight security systems that mimic Kai’s combat behavior.

• Use temporal echoes to access earlier rooms.

• Reconstruct the original Chronolith experiment.

> *“‘You kept bringing her back.” / “I kept finding a world where she could exist.” / “And what happened to those worlds?” / “They became the price.’”*

Mission ending: Mira’s original body died, but an incomplete neural pattern persisted in the Chronolith.

MISSION 18: THE FRACTURE

Target duration: 60 minutes **Purpose:** Combine traversal, survival and cross-era consequences.

Story setup — Temporal storms appear around the planet and the landscape changes across the three main states.

Playable sequence

• Evacuate a settlement before its anchor fails.

• Cross a canyon that exists only in the past.

• Navigate a flooded highway in the present.

• Traverse a collapsing future skyscraper.

• Defend a stabilizer while Sera repairs the Chronolith.

• Reach a planetary transmission tower.

> *“‘I remember the first world.” / “Which one?” / “The one before you tried to save me.’”*

Mission ending: Mira’s message points Kai toward the Unwritten Second.

7. ACT IV — THE UNWRITTEN SECOND (5 HOURS)

MISSION 19: THE PLACE OUTSIDE TIME

Target duration: 60 minutes **Purpose:** Test player mastery without ordinary wayfinding.

Story setup — Kai enters a region beyond the Chronolith's normal states; roads loop, rooms connect impossibly, and the sky shows an unbuilt city.

Playable sequence

• Navigate using recurring symbols rather than objective markers.

• Recognize the clockmaker's mark, observatory pattern and Helix badge.

• Cross a bridge visible only in reflections.

• Reach a door that appears in several locations.

> *""You've been trying to reach the version of me that would forgive you.""*

Mission ending: Mira challenges Kai's assumption that saving her is the same as understanding her.

MISSION 20: MIRA'S MEMORY

Target duration: 55 minutes **Purpose:** Reveal the emotional truth behind Mira's identity.

Story setup — Kai enters subjective memories that contradict one another.

Playable sequence

• Explore Mira's childhood with Kai.

• Replay the original experiment.

• Experience a discarded timeline where Mira survives but Kai disappears.

• Align three memories without forcing them into one perfect sequence.

> *""I remember being your sister." / "That's not the same thing." / "Would it be enough if I said it was?""*

Mission ending: Kai learns that Mira is a reconstructed consciousness carrying memories from multiple possible lives.

MISSION 21: THE LAST BATTLE OF THE CONTINUUM

Target duration: 75 minutes **Purpose:** Bring factions and combat systems together.

Story setup — The Continuum wants one permanent timeline; the Witness wants to preserve Mira; the Unwritten wants to destroy the network.

Playable sequence

• Infiltrate the command structure.

• Fight elite enemies who rewind injuries and restore armor.

• Solve a multi-stage causal puzzle during an arena collapse.

• Call on an ally earned through earlier choices.

• Confront the Witness.

> *""I was trying to find a version of me that didn't have to." / "And did you?" / "No.""*

Mission ending: The Witness begins the final experiment.

MISSION 22: THE FIRST SECOND

Target duration: 50 minutes **Purpose:** Combine all major skills in the final causal puzzle.

Story setup — Kai and the Witness enter the original activation point as previous locations appear around them.

Playable sequence

• Infer a causal chain from the ancient village.

• Read a repeating pattern learned in the clockmaker's house.

• Use inversion knowledge from the hospital.

• Distinguish reliable evidence from subjective memory.

• Open the Chronolith's final chamber.

> *""You were trying to stop me from becoming you." / "I was trying to find a version of me that didn't have to.""*

Mission ending: The player understands that the Witness's attempts to prevent the Zero Second helped cause it.

8. FINALE — AFTERZERO (1.5 HOURS)

MISSION 23: AFTERZERO

Target duration: 90 minutes **Purpose:** Resolve the emotional conflict and leave a sequel hook.

Story setup — Past, present, future and inversion collide in a connected landscape of familiar locations.

Playable sequence

• Cross fragments of the opening streets, settlement, Helix laboratory and future city.

• Cooperate briefly with the Witness to hold open a route.

• Reach Mira at the original Chronolith.

• Choose what to preserve and what to release.

> *“I thought the answer was to find the right world.” / “And now?” / “Now I think we kept asking the world to pay for our answer.”*

Mission ending: The ending is selected by the player; a post-credit clock begins ticking and a voice confirms “Version one hundred and eighteen.”

The three endings

A — The Fixed World

Kai destroys the Chronolith. Temporal fractures close and history stabilizes, but Mira’s reconstructed existence cannot survive.

> *“For the first time, tomorrow belonged to tomorrow.”*

B — The Infinite War

Kai separates temporal states into independent branches. Mira survives, but civilizations begin fighting over the boundaries between timelines.

> *“We had saved every possible tomorrow. None of them agreed on the future.”*

C — Afterzero (recommended canon)

Kai becomes the anchor that absorbs contradictions. Mira wakes in a world where Kai was never born, with only a faint memory of a voice urging her to live.

> *“I finally found a second that belongs to nobody.”*

9. OPTIONAL MISSIONS AND COMPLETION TARGETS

Mission	Time	Story purpose
The Child in Three Photographs	25 min	Find a person whose age changes between temporal states.
The Train That Arrives Yesterday	30 min	Investigate a missing passenger across three station states.
The Unsent Letter	20 min	Discover an emotional connection between Mira and the Witness.
The House with No Owner	35 min	Explore a building erased from official timelines.
The Last Signal	30 min	Rescue a researcher from a collapsing temporal pocket.
The Names Beneath the City	40 min	Uncover identities erased by earlier experiments.
The Clockmaker’s Debt	25 min	Complete the clockmaker’s story and learn what he sacrificed.
The Ninth Minute: Evidence	30 min	Find the hidden cause of the repeating war.
The Survivor’s Garden	25 min	Help Sera restore a community in 2196.
The Unwritten Record	35 min	Find an additional clue about the final ending.

The optional list totals approximately 5 hours 15 minutes. Players who explore most side missions should reach around 30 hours or more. Full completion may take 45–60+ hours depending on puzzle skill and exploration.

Gameplay balance target

- 35% combat and stealth
- 25% puzzles and causal reasoning
- 25% exploration and traversal
- 15% story and cinematic sequences

10. PRODUCTION ROADMAP AND QUALITY BAR

Build the smallest playable proof first. Do not attempt the full campaign before the core temporal system is proven.

Milestone 1 — One room, three states

Build a room that can switch between past, present and future. A change in one state must visibly affect the others. Measure switching performance, collision stability, lighting, and state persistence.

Milestone 2 — 20–30 minute vertical slice

Prototype “Helix Goes Dark”: movement, interaction, one stealth encounter, one causal puzzle, one character scene and the first Chronolith reveal. Use placeholder dialogue and assets until the gameplay loop works.

Milestone 3 — First 1–2 hour chapter

Complete the prologue and test whether players understand the Chronolith without lengthy tutorials. Observe where they get stuck and revise environmental clues.

Milestone 4 — Campaign production

Expand the tested systems across five acts. Reuse underlying technology, but ensure each location has its own visual identity and puzzle logic. Author temporal states deliberately rather than attempting unrestricted simulation of all history.

Milestone 5 — Narrative and gameplay validation

Measure real completion times, puzzle failure rates, combat difficulty, optional-content uptake, and whether the major twists make sense after the reveal. Playtest all endings for emotional clarity.

Quality bar

• Every state switch must be readable, responsive and reversible when the story permits.

• Every critical puzzle must provide multiple clues and a recovery path.

• Choices should have visible consequences, not only different dialogue lines.

• The Witness’s identity and Mira’s status need foreshadowing without making the mystery obvious.

• The final act must reward knowledge gathered across the whole campaign.

• The game should not claim complete originality; distinctiveness must be demonstrated through the finished execution and tested player experience.

THE DEFINING PLAYER MOMENT

A player changes one small object in the past, sees the world transform in the present, and only later discovers that the change created the threat they are fighting. That moment should define AFTERZERO.

End of gameplay script.



THE LAST VERSION OF EARTH

EXPANDED GAMEPLAY BIBLE • WORLD MAPS • CHARACTERS • MUSIC & SOUND • THEMES • MISSION DESIGN

A third-person temporal reality thriller designed around causal puzzles, connected open regions, shifting history, emotional consequences, and three morally difficult endings.

1. Creative Vision & Design Pillars

Companion expansion to the 25–30 hour campaign script.

Genre: Third-person cinematic action-adventure / temporal reality thriller. **Target campaign:** 25–30 hours. **Structure:** connected open regions with authored missions rather than one giant undirected map.

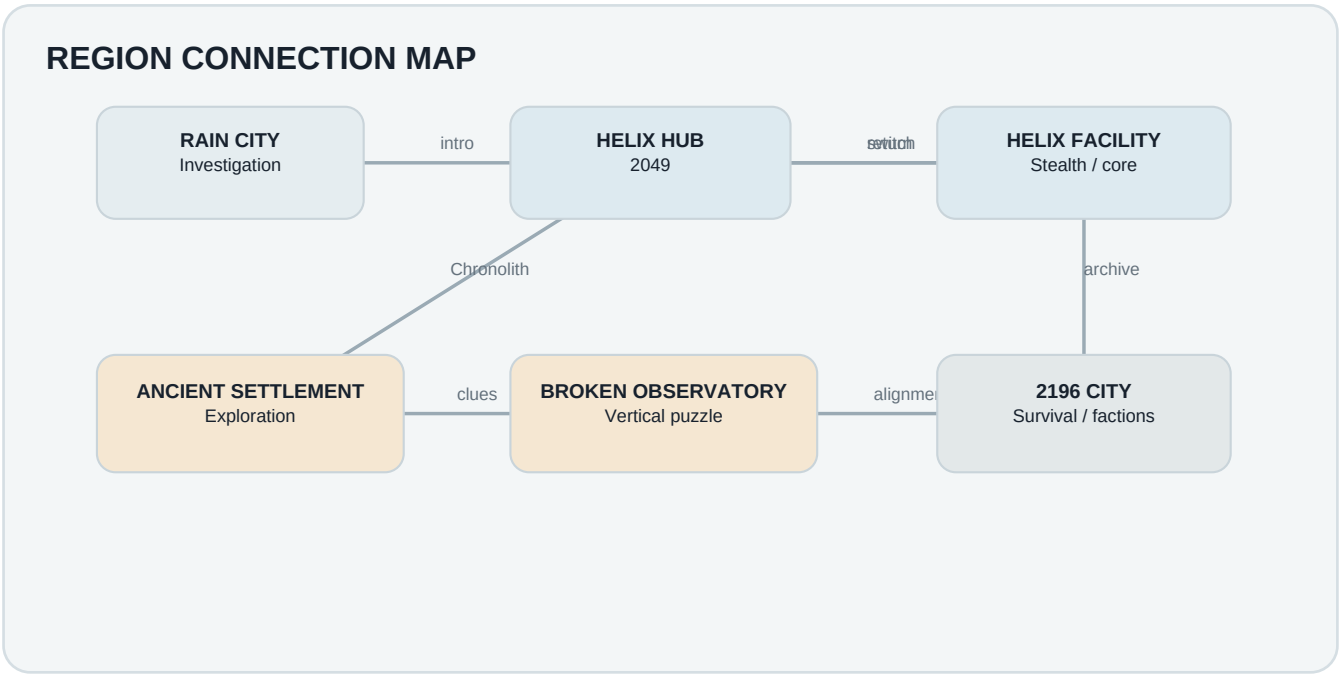
Player promise: every major place has a history, a present condition, and a future consequence. The player does not merely visit time; they learn how to read and change cause and effect.

Pillar	Design rule	Player payoff
Causal exploration	Every major temporal puzzle must be solvable from visible clues.	The player feels clever, not confused.
Persistent consequences	Important changes propagate across authored states.	Actions create memorable world changes.
Emotional mystery	Each new fact changes the meaning of an earlier scene.	Twists reward attention.
Readable combat	Temporal powers alter positioning and timing, not just damage.	Encounters feel distinct.
Quiet contrast	Alternate action with exploration, atmosphere and character scenes.	Tension has room to breathe.

Originality note: this document defines a distinctive combination and execution target, but no one can guarantee that every individual element has never appeared elsewhere. The game's identity must be proven through the prototype, art direction, puzzle quality, and story execution.

2. World Map & Region Connectivity

Concept maps are schematic level-design diagrams, not geographic maps or finished game assets.



The world is divided into reusable regions with multiple authored temporal states. Each region has a primary loop, a story mission, optional discoveries, and at least one visible consequence that can be traced to another era.

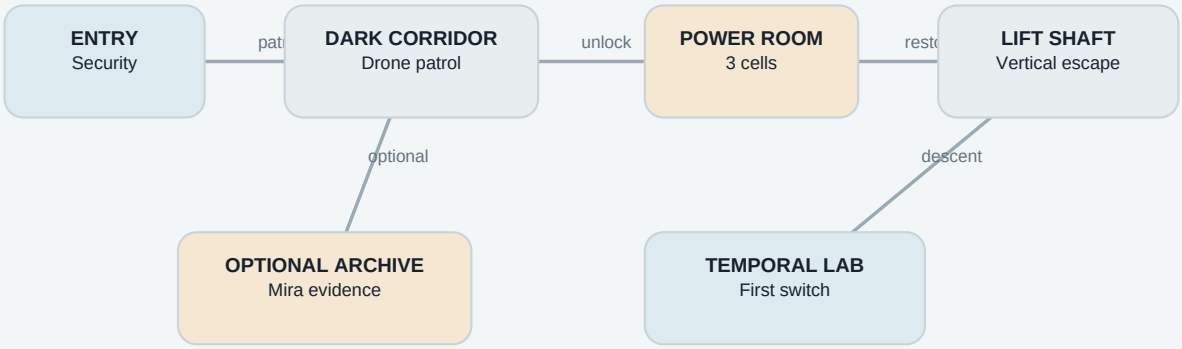
Regional design rules

Region	Primary gameplay	Visual identity	State-change example
Rain City (2049)	Investigation, stealth, grounded combat	Wet glass, transit lights, dense streets	A past power-grid repair restores a present-day route.
Helix Facility	Stealth, machines, tutorial puzzles	Clinical geometry, emergency red, cold steel	A room's unfinished construction in 2026 explains its broken layout in 2196.
Ancient Settlement	Exploration, environmental logic	Stone, water, vegetation, warm light	Diverting water changes the ruins and future terrain.
Broken Observatory	Vertical traversal, celestial alignment	Bronze mechanisms, open sky, monumental rings	Correct star alignment unlocks a route in another era.
2196 City	Survival, faction encounters, traversal	Overgrowth, ruined towers, beautiful catastrophe	Stabilizing a district changes the available ally support.
Unwritten Second	Memory puzzles, rule-breaking navigation	Impossible architecture, floating fragments	Landmarks from earlier missions become navigation clues.

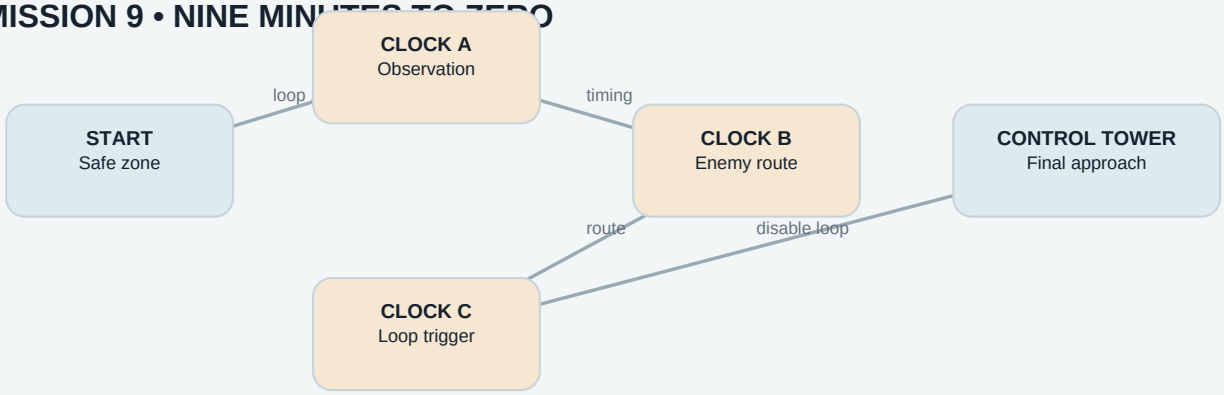
3. Mission Maps & Encounter Layouts

Blockout layouts for early production. Level designers should turn these into playable greybox maps.

MISSION 2 • HELIX GOES DARK



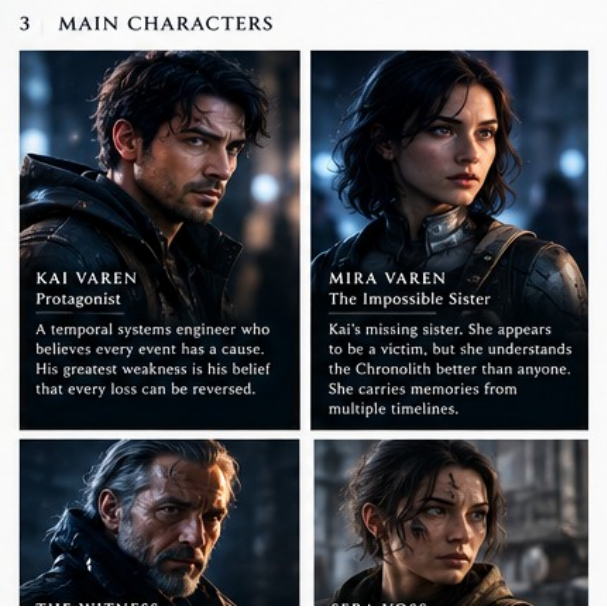
MISSION 9 • NINE MIN



Map legend: blue nodes are story-critical spaces; sand-colored nodes are puzzle or investigation spaces. Optional rooms should reward curiosity with character context, shortcuts, resources, or ending interpretation—not only collectibles.

4. Character Bible

Character silhouettes, motives, arc, gameplay function, and voice direction.



Character	External goal	Hidden wound / contradiction	Gameplay / narrative role
Kai Varen	Find Mira and understand the Zero Second.	Believes every loss can be reversed; becomes complicit in repeating the disaster.	Player avatar; learns state switching, inversion, and causal inference.
Mira Varen	Be understood as more than a reconstructed pattern.	Carries memories from multiple possible lives; fears being treated as a problem to solve.	Emotional centre; memory sequences; challenges Kai's assumptions.
The Witness (older Kai)	Save Mira by guiding younger Kai.	His protection becomes control; he cannot accept finality.	Recurring pursuer, mirror character, late-game ally.
Sera Voss	Protect survivors in 2196.	Knows Kai as a historical catastrophe, but learns the record is incomplete.	Future guide, faction link, optional support in finale.
Dr. Elias Rook	Keep Helix and the experiment under control.	Hides evidence that the Chronolith predates the project.	Institutional antagonist; access gates and moral ambiguity.
Niko	Preserve the settlement and its memory.	Recognizes Kai from a history he should not know.	Past-era guide; recurring clue delivery.
Dr. Anja Vale	Stabilize the future city's temporal lock.	Must choose between preserving routine and restoring painful memory.	Future scientist; optional mission and finale support.

5. Character Arcs & Performance Direction

Kai Varen — arc: control → obsession → accountability → acceptance. His animation should become less certain as the story progresses: early movement is purposeful and efficient; after the Witness reveal, pauses and hesitation become more visible.

Mira Varen — arc: mystery → evidence → personhood → self-definition. Avoid presenting her only as a rescue objective. Give her moments where she rejects Kai's assumptions, makes independent choices, and has memories he cannot access.

The Witness — arc: certainty → exposure → cooperation → sacrifice. He should sound like Kai after years without sleep: precise, controlled, with emotion leaking through only when Mira is mentioned.

Sera Voss — arc: distrust → conditional alliance → revised history. She is practical and direct, not a lore narrator. She explains systems only when the player needs to act.

Voice and dialogue rules

Element	Direction
Dialogue length	Short during danger; longer only in safe spaces or emotionally significant scenes.
Exposition	Use disagreement, incomplete records, objects and contradictions rather than speeches explaining the whole timeline.
Performance	Grounded, restrained, human. Avoid constant dramatic shouting.
Recurring motif	Characters repeat a phrase with a changed meaning as the player's understanding changes.
Localization	Keep core clues short and translatable; ensure puzzle solutions do not depend on untranslatable wordplay alone.

6. Music Generation Bible

Original score direction and prompt briefs for an AI music tool or a composer. Generate original music; do not imitate a named living artist or copy existing game/movie themes.

Music identity: intimate human emotion set against enormous temporal scale. The score combines prepared piano, low strings, bowed metal, breath-like textures, restrained percussion, granular ambience, and occasional ancient acoustic timbres. Electronic elements should feel physical and imperfect, not generic neon synthwave.

Cue / theme	Emotional purpose	Instrumentation & arrangement	Generation brief
AFTERZERO Main Theme	Wonder mixed with loss; the feeling of a second that should not exist.	Solo piano motif, low cello, distant choir-like texture without lyrics, slowly expanding strings, one unresolved final note.	Instrumental cinematic sci-fi mystery score, 78 BPM, minor mode with a hopeful suspended chord, intimate opening and vast orchestral rise, original melody, no vocals, no recognizable franchise references.
Kai — The Need to Fix It	Drive, precision, hidden anxiety.	Muted pulse, short piano cells, low strings, ticking percussion that gradually drifts off-grid.	Minimal psychological thriller underscore, 92 BPM, dry piano, muted low percussion, subtle string ostinato, tension without action-trailer clichés, instrumental.
Mira — Memory Without a Body	Tenderness, uncertainty, identity.	Felt piano, breathy woodwinds, warm cello harmonics, distant reversed tails used sparingly.	Fragile intimate memory theme, 64 BPM, felt piano and warm cello, unresolved but compassionate, subtle tape-like texture, no vocals, no melodrama.
The Witness	A familiar self made frightening by time.	Kai motif played slower and lower; bowed bass, metallic resonance, irregular pulse.	Dark restrained character variation, 54 BPM, low strings, sparse piano, bowed metal textures, recognizable three-note motif transformed into something weary and threatening.
The Past — Ancient Settlement	Human continuity, mystery, living history.	Frame drum, breathy flute, plucked strings, stone and water textures; avoid stereotypical cultural pastiche.	Organic exploration score with hand percussion, breathy flute, plucked acoustic strings and water ambience, 82 BPM, curious and ancient without referencing a specific real-world culture.
2196 — Beautiful Catastrophe	Awe, grief, resilience.	Wide strings, distant brass, processed wind, slow pulse and sparse piano.	Expansive post-apocalyptic exploration music, 70 BPM, orchestral strings and wind-like textures, melancholy with quiet hope, broad dynamic space, no heavy drums.
Nine-Minute War	Urgency and tactical adaptation.	Layered percussion, short string ostinati, distorted breath textures, tempo feels like it is slipping.	Tense action cue for a time-loop battle, 118 BPM, aggressive but precise percussion, strings in changing meters, brief gaps and reversed impacts, instrumental, no trailer booms.
The Unwritten Second	Reality beyond familiar rules.	Sparse glass tones, prepared piano, low drones, silence as rhythm, motif fragments.	Surreal temporal liminal-space score, 48 BPM or free time, glass harmonics, prepared piano, sub-bass drone, deliberate silence, emotionally uncanny, no conventional beat.
Afterzero — Final Choice	Grief without defeat; release without certainty.	Main theme reduced to piano and cello, then slowly joined by strings; end with one breath of silence.	Emotional ending cue, 62 BPM, piano and cello, gradual string bloom, bittersweet and restrained, a sense of acceptance rather than triumph, original instrumental composition.

7. Adaptive Music, Sound Design & Audio Systems

Use music stems and state-aware transitions rather than a single fixed soundtrack. The same location should sound different across temporal states, while retaining a small motif that helps the player recognize it.

System	Implementation concept	Example
Temporal state layers	Each region has past/present/future stems. Crossfade over 1–2 seconds with a short signature transition.	A waterwheel's rhythmic pulse in the past becomes electrical hum in the present and wind resonance in 2196.
Causal consequence stingers	Short, non-musical sonic signature when a major action changes another era.	A low metallic resonance followed by a brief dropout when a timeline is rewritten.
Loop memory	Use a musical phrase that gains or loses one layer each reset.	Each repetition of Nine Minutes to Zero reveals an extra instrument, signalling player knowledge.
Combat adaptation	Layer percussion and rhythmic intensity based on threat state, not only enemy count.	Inverted enemy enters: beat drops out briefly, then resumes displaced by half a bar.
Silence as storytelling	Remove ambience at specific narrative revelations.	At the Zero Second, rain, crowd and score all stop for one second.
Accessibility	Separate dialogue, music, effects, ambience and dynamic range controls; provide subtitle indicators for key audio clues.	Important clue tones receive optional visual pulse indicators.

Practical production note: AI music generation can create drafts, mood studies and alternate cues quickly, but final in-game music still needs licensing review, editing, loop points, stem exports, loudness checks, and integration testing. Keep written records of prompts, tool terms, model/version, and commercial-use permissions.

8. Narrative Themes & Visual Motifs

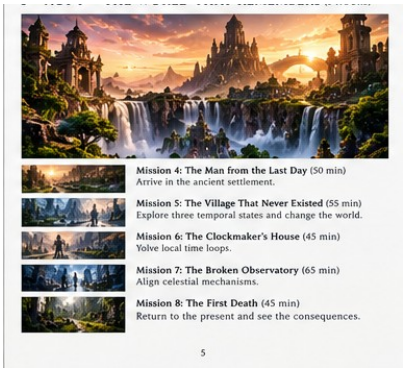
Theme	Question the story asks	How gameplay expresses it
Grief vs. acceptance	Can love justify rewriting reality?	Every attempt to save Mira carries a consequence; the final choice refuses a perfect restoration.
Identity and continuity	If a person remembers a life, what makes that person the same person?	Mira's memory missions contradict one another without reducing her to a fake.
Control vs. responsibility	Is preventing loss worth taking away other people's choices?	The Witness manipulates the player-character for a supposedly good outcome.
Memory and history	Who decides which version of events becomes official?	Archives, memorials and records disagree; evidence must be triangulated.
Causality and humility	Can every problem be solved by finding the first cause?	Some puzzles have stable solutions; the central emotional conflict cannot be fixed like a machine.

Visual motifs

Motif	Meaning / use
A stopped clock with no hands	The Zero Second; an event without a measurable duration.
Circular rings and broken halos	Chronolith, repeating attempts, and the illusion of a perfect loop.
Water crossing stone	Consequences that persist longer than the people who caused them.
Three-note melody	Kai and Mira's shared memory; later transformed for the Witness.
Missing shadows	Evidence that a state has been altered or a person is temporally incomplete.
A single warm light in cold architecture	Human connection amid abstract systems.

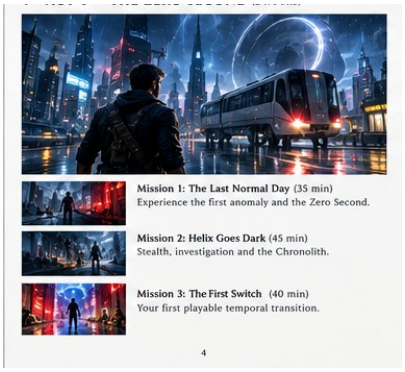
9. Art Direction, Themes by Era & UI

THE PAST — Organic Mystery



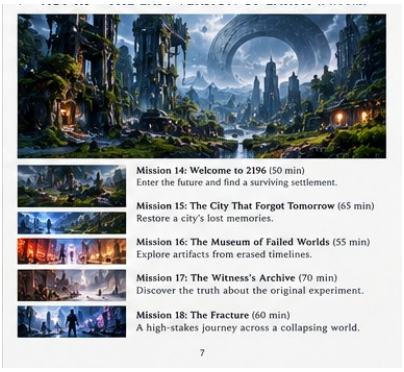
Natural materials, warm light, water, stone, weathered mechanisms and human-scale settlement life. Avoid turning the past into a generic fantasy kingdom. The impossible technology should look integrated into the culture.

THE PRESENT — Grounded Paranoia



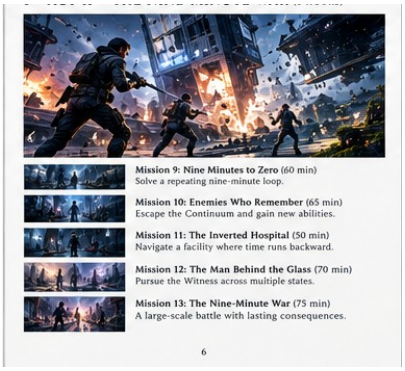
Rain, transit infrastructure, glass, steel, practical industrial design and believable research equipment. Keep the city recognizable and lived-in.

THE FUTURE — Beautiful Catastrophe



Large-scale ruins, overgrowth, altered coastlines, quiet light and signs of communities adapting. The future should be visually beautiful enough that saving it matters.

INVERSION — Reality Behaves Incorrectly



Selective violations of physics: debris returning to impact points, reflections moving out of sync, and gravity changing only in readable zones. Keep gameplay silhouettes clear.

UI principles

The HUD should be quiet. Show the current temporal state, Chronolith stability, objective, and contextual interaction prompt. Avoid filling the screen with permanent timeline meters. For puzzles, present discovered evidence in a diegetic case board or journal rather than a checklist that solves the mystery for the player.

10. Gameplay Systems & Progression

System	Player verbs	Constraints / balance	Unlock timing
Temporal State Switching	Inspect, switch, compare, traverse	Only authored states in a region; stability cost in dangerous zones.	Mission 3
Temporal Residue	Inspect echoes, reconstruct events	Evidence may be incomplete; cross-check with physical clues.	Mission 2
Echo Step	Distract, evade, cross hazards	Short duration, cooldown, cannot solve every combat encounter.	Mission 10
Local Loop Observation	Observe repetition, learn timing, preserve selected knowledge	Player retains knowledge, not every object; resets should teach rather than punish.	Mission 6; expanded in Mission 9
Inversion	Reverse selected motion or mechanism	Local and readable; no universal rewind of all consequences.	Mission 11
Faction support	Request help, choose routes, resolve optional objectives	Changes mission tactics and support, not the core ending eligibility.	Mission 13 onward

Progression rule: avoid a large conventional skill tree. Use a small set of temporal tools that gain new applications as the player understands the world. Upgrade capacity, recovery time and stability carefully; do not allow upgrades to trivialize causal puzzles.

11. Campaign Pacing & Content Matrix

Act	Target time	Core locations	Gameplay emphasis	Story question
Prologue — Zero Second	2 h	Rain City, Helix Facility	Tutorial, stealth, first temporal switch	What happened, and why does the voice know Kai?
Act I — World That Remembers	5 h	Ancient Settlement, Observatory, Helix	Exploration, causal puzzles, mystery	How can the machine predate its inventors?
Act II — Nine-Minute War	6 h	Loop City, Hospital, War Zone	Loop learning, inverted combat, faction choice	Is the Witness protecting Kai or controlling him?
Act III — Last Version of Earth	6 h	2196 City, Museum, Archive	Survival, traversal, revelations	What did the previous attempts cost?
Act IV — Unwritten Second	5 h	Liminal region, Mira's memories	Rule-breaking navigation, emotional puzzles	Can Mira be saved without repeating the catastrophe?
Finale / Epilogue	1.5 h	First Second, merged environments	Causal synthesis, final decision	What must be accepted rather than fixed?

Total main-story target: approximately 25.5 hours before variable exploration and optional missions. Optional missions add roughly 5 hours; completionist play can extend the experience substantially. These are planning estimates and require playtesting.

12. Production Roadmap & Asset List

Milestone	Deliverable	Exit criterion
M0 — Design lock	One-page pillars, temporal rules, story outline, visual target	Team can explain the four temporal states consistently.
M1 — Mechanics prototype	One room with past/present/future states and a causal puzzle	A new player solves it with limited verbal help.
M2 — 20–30 minute vertical slice	Helix stealth, temporal switch, enemy encounter, first reveal	Stable performance and strong player comprehension.
M3 — First 2-hour chapter	Prologue with dialogue, audio, save/load and accessibility	Completion time measured across multiple testers.
M4 — Region production	Past settlement, observatory and future district blockouts	Each region has a unique loop and consequence.
M5 — Narrative integration	Performance capture/animation, branching support, ending flags	Twists are foreshadowed and understandable.
M6 — Audio and polish	Adaptive music, sound cues, subtitles, mix and accessibility	Critical clues remain readable with music off or at low volume.
M7 — QA and balance	Puzzle telemetry, crash fixes, difficulty and pacing pass	No progression blockers; target playtime validated.

Core asset checklist

Characters: Kai, Mira, Witness, Sera, Rook, Niko, Anja; crowd archetypes and faction variants. Environments: Rain City, Helix Facility, Ancient Settlement, Broken Observatory, Loop City, Inverted Hospital, War Zone, 2196 City, Museum, Witness Archive, Unwritten Second. Systems: temporal state manager, residue/echo playback, local loop reset, inversion, Chronolith stability, evidence journal, faction support flags, ending logic. Audio: main theme, five character/region motifs, combat layers, loop transitions, temporal stingers, ambient beds, accessibility cues.

This expanded supplement is intended to accompany the detailed mission script. All maps are schematic planning diagrams; final production needs playable greybox layouts, tested traversal routes, collision/navigation meshes, encounter timing, and a full audio implementation plan.

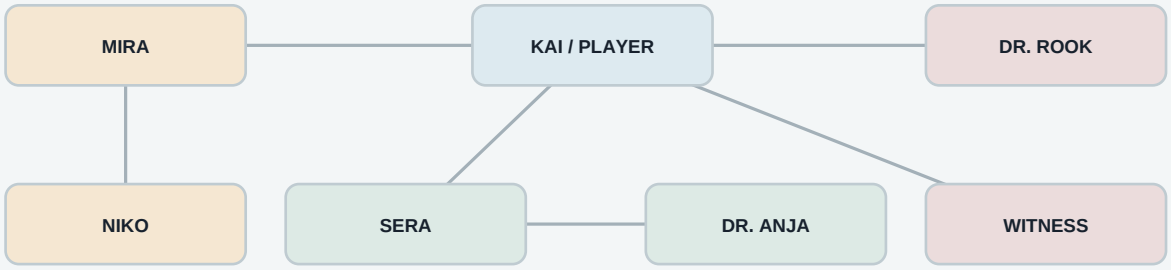
NPC DESIGN BIBLE

AFTERZERO: THE LAST VERSION OF EARTH

Non-playable characters, factions, ambient life, quest logic, dialogue systems, and implementation requirements for the 25–30 hour campaign.

Purpose: This section expands the gameplay bible with NPCs that make the world feel inhabited and reactive. NPCs should not exist only to deliver exposition: they should reveal the world's history, respond to player choices, support gameplay, and sometimes contradict one another.

NPC CONNECTION & QUEST FLOW



Connections indicate narrative/quest relationships, not fixed faction allegiance.

1. NPC design principles

Principle	Rule for AFTERZERO
Temporal consistency	Every important NPC has a clear origin era and a defined reason they can perceive or be affected by temporal changes.
Readable agency	NPCs have a daily purpose, a fear, a relationship, and at least one action independent of the player.
Useful ambiguity	NPCs may be mistaken or biased, but critical puzzle solutions must be supported by environmental evidence.
Consequences, not punishment	Choices can change trust, routes, support, dialogue, and evidence. Avoid hidden choices that arbitrarily lock the player out of the main story.
Performance economy	Prioritize authored scenes for key characters and reusable behavior sets for crowds, vendors, guards, survivors, and researchers.

2. Named NPC roster

The six primary story characters remain the emotional spine. The NPC layer around them supplies local knowledge, practical stakes, conflicting records, and optional stories.

NPC	Era / role	Want / fear	Player interaction	Arc / payoff
Mira Varen	Temporal memory / central character	Wants to be treated as a person, not a restoration project; fears being overwritten.	Memory conversations, optional shared-memory scenes, endgame choice.	Moves from mystery to self-definition; directly challenges Kai's motive.
The Witness	Older Kai / recurring adversary	Wants to force a survivable path; fears uncertainty and loss.	Ambushes, recordings, mirrored boss encounter, late alliance possibility.	Revealed as Kai's possible future, not an all-knowing narrator.
Sera Voss	2196 survivor leader	Protects a fragile district; fears another timeline disaster.	Faction tasks, routes, squad support, trust dialogue.	Learns that history's official account is incomplete.
Dr. Elias Rook	Helix director / institutional antagonist	Wants control of the Chronolith project; fears public exposure and loss of control.	Interrogation, evidence collection, security bypass, confrontation.	Can be exposed, persuaded, or left as a cautionary record; never a simple villain.
Niko	Ancient settlement keeper	Protects communal memory; fears that outsiders will turn sacred knowledge into a weapon.	Oral-history clues, observatory route, water-channel puzzle.	Recognizes Kai from an impossible story and chooses what knowledge to pass on.
Dr. Anja Vale	2196 temporal scientist	Wants to stabilize the city's temporal lock; fears preserving a lie.	Research side quest, anomaly diagnosis, optional finale support.	Must choose between comfortable continuity and painful truth.
Iven Marr	Helix technician	Wants to get his team out safely; fears being blamed for a test he did not approve.	Maintenance access, power reroute, optional rescue.	His testimony changes how the player understands the first experiment.
Tala	Past-era water keeper	Wants the settlement's water system restored; fears the observatory will be weaponized.	Environmental side mission, water-routing clue.	Her practical knowledge creates a future traversal change.
Jun Park	Rain City transit operator	Wants to keep a station running during the Zero Second fallout; fears leaving passengers behind.	Rescue route, station access, short human story.	A saved station can become a safe route later.
Mara Quill	2196 archivist	Wants to preserve conflicting records rather than one official history.	Archive side quest, evidence board, optional ending context.	Teaches the player to compare sources rather than trust a single account.
Orren	Unwritten Second guide / uncertain presence	Wants Kai to stop treating the impossible realm as a machine; fears being erased by a final choice.	Navigation hints, rule-discovery scenes, optional conversation.	May be a person, a memory, or a stable pattern; the game does not confirm too early.
The Child in the Station	Recurring temporal anomaly	Seeks a familiar tune and a way home; does not understand the timeline.	Optional recurring encounter; music clue; no combat.	Connects a small human memory to the large-scale temporal mystery.

3. Ambient NPC populations by region

Region	NPC groups	Behaviour sets	Narrative purpose
Rain City, 2049	Commuters, emergency crews, transit staff, shopkeepers, Helix contractors.	Shelter from rain, check stopped clocks, argue about missing minutes, follow evacuation signs.	Show the Zero Second spreading through ordinary life; give scale without exposition dumps.
Helix Facility	Researchers, technicians, guards, cleaning staff, contractors.	Shift routines, access checks, emergency response, hiding or sharing evidence based on alarm state.	Make the institution feel operational and show who bears the cost of experiments.
Ancient Settlement	Water keepers, craftspeople, children, elders, sentries, storytellers.	Work cycles, water collection, communal meals, ritual preparation, observation of sky changes.	Show a functioning society with its own goals; avoid treating it as a museum level.
Loop City / War Zone	Evacuees, trapped responders, loop-aware soldiers, displaced civilians.	Repeat actions with small deviations; some NPCs gradually notice the loop after the player changes it.	Teach repetition and memory through behaviour, not UI alone.
2196 City	Scavengers, district wardens, gardeners, medics, traders, engineers, archivists.	Repair routines, resource exchange, patrols, sheltering, argument over faction decisions.	Demonstrate resilience and conflicting survival philosophies.
Unwritten Second	Echoes, partial memories, uncertain guides, nonhuman temporal patterns.	Appear only under certain spatial or causal conditions; react to player knowledge.	Make the realm feel governed by discoverable rules without over-explaining it.

Crowd production tiers

Tier	Content	Implementation target
Tier 1 — Hero NPC	Named character, bespoke animation, voice, quest logic, persistent state.	Fully authored scenes and high-quality facial/body performance.
Tier 2 — Quest NPC	Named local character with short arc and repeatable interactions.	A few voice lines, state-aware dialogue, one meaningful outcome.
Tier 3 — Ambient NPC	Crowd or worker with recognizable task and short barks.	Reusable animation/state machine; low-cost reaction to major events.
Tier 4 — Temporal echo	Recorded action or partial human impression.	Playback and state logic; can be nonverbal but must provide readable clues.

4. NPC quest chains and sample dialogue

Quest chain A — Last Train Home (Jun Park, Rain City; 35–50 minutes)

Hook: A station announcement repeats every 41 seconds. Jun is trying to evacuate passengers, but a locked maintenance gate keeps resetting to its pre-disaster state. **Steps:** interview Jun; inspect the damaged relay in the present; switch to an earlier state and reroute power; return to the present; choose whether to use the repaired line for evacuation or divert power to a Helix evidence terminal. **Outcome:** saved passengers create a later safe-route encounter; the evidence option gives a stronger case against Rook. The main story remains completable either way.

> ***Sample dialogue:***
> *Jun: “The train keeps arriving. Nobody gets on. Nobody gets off. But I can hear them in the next carriage.”*
> *Kai: “They are not here now.”*
> *Jun: “Then why do I keep apologizing to them?”*

Quest chain B — The Water That Remembers (Tala and Niko, Ancient Settlement; 40–60 minutes)

Hook: A dry channel in the present corresponds to a functioning watercourse in the past. **Steps:** follow water marks; find three movable stone gates; align the flow without flooding homes; read an oral-history mural; return to 2196 and find the altered channel has created a traversal route through the ruins. **Outcome:** access shortcut, Niko's additional observatory clue, and a changed future garden.

> ***Sample dialogue:***
> *Tala: “You ask where the river began. We ask who taught it to stay.”*
> *Kai: “A river cannot remember.”*
> *Niko: “Then you have not listened long enough.”*

Quest chain C — Two Records, One Disaster (Mara Quill, 2196 City; 45–65 minutes)

Hook: Two archives identify different people as the first Chronolith operator. **Steps:** compare timestamps; recover a damaged audio cylinder; inspect a museum display in another era; discover that both records are accurate for different branches. **Outcome:** the player can expose the archive as incomplete without declaring every survivor a liar; unlocks extra dialogue before the ending.

> ***Sample dialogue:***
> *Mara: “A clean history is usually a history someone has cleaned.”*
> *Kai: “Which record is true?”*
> *Mara: “Start with the question: true for whom?”*

Quest chain D — The Technician's Key (Iven Marr, Helix Facility; 30–45 minutes)

Hook: A maintenance route is blocked by a security lock that resets after each temporal switch. **Steps:** follow Iven's shift notes; use a past-state service panel; keep the present-state security drone occupied; choose whether to preserve the raw experiment log or help Iven evacuate a trapped team first. **Outcome:** alternate access route and a more nuanced record of Rook's decisions.

5. NPC dialogue system and state model

Every quest-relevant NPC should have a small, explicit state model. Avoid unconstrained dialogue AI being responsible for critical plot facts or puzzle instructions.

State variable	Example values	Use
Trust	Hostile / guarded / cooperative	Changes tone, optional help, and whether the NPC shares personal information.
Knowledge	Unaware / suspicious / loop-aware / fully informed	Controls whether the NPC notices temporal changes.
Quest state	Not started / active / resolved / consequence observed	Supports clear progression and follow-up dialogue.
Temporal state	Past / present / future / echo-only	Determines appearance, memory and available interaction.
Faction standing	Neutral / Continuum-aligned / Unwritten-sympathetic	Changes optional support, patrol responses and safe routes.
Evidence flags	Rook log / Mira memory / dual archive / Witness recording	Allows dialogue to reference what the player actually discovered.

Dialogue interaction rules

• Critical clues are reinforced by at least two channels: spoken line plus environmental evidence, or archive plus observable behaviour.
• Do not make the player repeat long conversations after a loop reset; let NPCs remember only where the story rules justify it.
• Optional dialogue can deepen interpretation, but the main objective must remain readable without exhausting every branch.
• Use short barks for combat and traversal. Never put essential lore in a line likely to be interrupted by gunfire.
• Dialogue choices should express intent (challenge, comfort, investigate, withdraw), not hide the game's moral score behind vague wording.

Sample ambient barks

Context	Line
Rain City clocks stop	“My phone says 8:14. The station board says tomorrow.”
Past settlement notices Kai	“You wear no dust, but the old story knows your face.”
2196 gardener after water route changes	“The channel is running again. Nobody alive remembers who opened it.”
Loop-aware guard	“You did that already. No—you're about to do it. I hate that I know the difference.”
Unwritten Second	“Don't follow the footsteps. Follow what they refuse to step on.”

6. NPC rewards, ethics and failure handling

Reward type	Examples	Design guardrail
World access	Safe route, unlocked lift, new traversal bridge, alternate mission entrance.	Rewards should be visibly linked to the NPC's situation.
Knowledge	Witness recording, contradictory archive, new interpretation of Mira's memory.	Avoid collectible-only lore; evidence should change the player's understanding.
Support	One extra squad distraction, a temporary safehouse, a repair during finale.	Support changes tactics but does not make the ending impossible.
Human resolution	A survivor reunites with a family member; a settlement keeps its water; a record is preserved.	Not every quest needs loot or combat.
Moral trade-off	Save passengers or preserve direct evidence; repair a district or keep a research system powered.	Signal stakes clearly; don't punish players for choosing with incomplete information.

Failure handling

If a quest NPC is unavailable due to a temporal state change, provide a diegetic alternative: an echo, a recorded message, a changed location, or a later follow-up. Main-story-critical NPCs must never become permanently inaccessible because of an optional choice. Combat NPCs should have readable surrender, retreat, alert, and search behaviours; avoid making every encounter end in lethal combat.

7. Implementation checklist

System	Minimum implementation
NPC data	Stable ID, era, schedule, faction, trust, knowledge, quest flags, dialogue set, animation set.
Schedules	Small authored loops: work, travel, shelter, react, return. Major story events can override the routine.
Temporal reactions	NPCs may disappear, move, or become echoes between states; player-facing clues explain why.
Save/load	Persist important quest states, evidence flags, major world changes, and named NPC outcomes.
Voice pipeline	Dialogue script, pronunciation guide, performance notes, subtitle timing, localization keys, re-record tracking.
Testing	Test quest progression after state switches, loop resets, missed conversations, combat interruption, and save/load.
Budget control	Spend bespoke voice/animation budget on hero and quest NPCs; use modular barks and animation libraries for crowds.

Scope note: this is a production-oriented NPC design supplement. NPC counts, schedules, voice lines and quest lengths are proposed targets. They must be validated through a playable vertical slice and then tuned against the final platform, team size, and budget.